

Rule 1: If a patient presents to the ED with possible STEMI, send passive alerts of a potential code STEMI to cardiology, cath lab, and pharmacy.

The screenshot displays the OpenEMR web interface. At the top, the patient's name is **CHERYL ADAMS (322)** with a dropdown arrow, and their DOB is 1972-02-11, Age 52. A search bar on the right says "Search by any demograph". Below the patient header, a breadcrumb trail shows: **Calendar** > **Clinical Reminders** > **2024-12-16 Encounter** > **Rule Detail Back**. The main content area is titled **patient presents w/ possible STEMI (Passive Alert)**. It contains several sections:

- Bibliographic Citations:** Includes fields for Developer, Funding Sources, Release, Web References, and Referential CDS (codetype:code).
- Reminder intervals (add):** A table with columns **Type** and **Detail**.

Type	Detail
Clinical	Warning: 1 Weeks, Past due: 1 Weeks
Patient	Warning: 1 Weeks, Past due: 1 Weeks
- Demographics filter criteria (add):** A table with columns **Criteria**, **Characteristics**, and **Requirements**.

Criteria	Characteristics	Requirements
(edit) (delete) v_fm_encounter_pari.class_code	Required Inclusion	Value: EMER Frequency: >= 1
(edit) (delete) Diagnosis	Required Inclusion	ICD10I21.9
- Target/Action Groups:** A section for **Clinical targets (add)** with columns **Criteria**, **Characteristics**, and **Requirements**.

Criteria	Characteristics	Requirements
(edit) (delete) Response Needed - Possible STEMI Patient	Required Inclusion	
Completed: Yes Frequency: >= 1 Interval: 1 x Months		

Create a new encounter for the patient with the class "Emergency Department" to indicate that the patient has presented to the ED.

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https://hanktest/openemr/interface/main/tabs/main.php?token_main=VQsGvJXO1wnPHpOCukWygXT6zQidCf1z7auQF6

Calendar Finder Flow Recalls Messages Patient Fees Modules Procedures Admin Reports Miscellaneous Popups

Search by any demographic

CHERYL ADAMS (322) ×

DOB: 1972-02-11 Age: 52

Select Encounter (0) +

Portal

Calendar Dashboard Patient Encounter

Visit Details

Visit Category: Encounter for problem

Class: Emergency Dept

Type: --Select One--

Sensitivity: normal

Encounter Provider: Administrator, Administrator

Referring Provider: -- Please Select --

Facility: Your Clinic Name Here

Billing Facility: Your Clinic Name Here

Date of Service: 2024-12-16 20:39

Onset/hosp. date:

In Collection: No

Discharge Disposition: -- Select One --

Reason for Visit

possible STEMI

Link/Add Issues to This Visit

+ Add Issue

A: aspirin
A: 2022-11-01 22:15:00 sulfa
A: 2023-04-02 00:00:00 pcn
P: Tension headache

To link this encounter/consult to an existing issue, click the desired issue

Create a medical problem with the ICD code for myocardial infarction to mimic a "possible STEMI" patient.

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Add/Edit Issue

Select Codes

ICD10 Diagnosis

Include Inactive

Ok Cancel

Show 15 entries

Search for: acute myocardial

Code	Description
I21.9	Acute myocardial infarction, unspecified
I23.0	Hemopericardium as current complication following acute myocardial infarction
I23.1	Atrial septal defect as current complication following acute myocardial infarction
I23.2	Ventricular septal defect as current complication following acute myocardial infarction
I23.3	Rupture of cardiac wall without hemopericardium as current complication following acute myocardial infarction
I23.4	Rupture of chordae tendineae as current complication following acute myocardial infarction
I23.5	Rupture of papillary muscle as current complication following acute myocardial infarction
I23.6	Thrombosis of atrium, auricular appendage, and ventricle as current complications following acute myocardial infarction
I23.8	Other current complications following acute myocardial infarction
I24.0	Acute coronary thrombosis not resulting in myocardial infarction

Showing 1 to 10 of 10 entries (filtered from 72,748 total entries)

Previous 1 Next

OpenEMR interface showing the Medical Record Dashboard for CHERYL ADAMS (322). The dashboard includes sections for Allergies, Medical Problems, Medications, Prescriptions, Demographics, and Patient Portal / API Access. The Medical Problems section lists conditions such as Tension headache, Chronic obstructive pulmonary disease, unspecified, Elevated blood pressure reading without diagnosis of hypertension, Unspecified essential hypertension, Acute myocardial infarction, unspecified, Conduct disorder, childhood-onset type, Reiter's disease, multiple sites, Pulmonary alveolar microlithiasis, and Drug or chemical induced diabetes mellitus with kidney complications.

Medical Record Dashboard - CHERYL M. ADAMS

Dashboard History Report Documents Transactions Issues Ledger External Data

Allergies Medical Problems Medications Prescriptions Demographics Patient Portal / API Access

Medical Problems:

- Tension headache
- Chronic obstructive pulmonary disease, unspecified
- Elevated blood pressure reading without diagnosis of hypertension
- Unspecified essential hypertension
- Acute myocardial infarction, unspecified
- Conduct disorder, childhood-onset type
- Reiter's disease, multiple sites
- Pulmonary alveolar microlithiasis
- Drug or chemical induced diabetes mellitus with kidney complications

An alert is triggered to cardiology, the cath lab, and pharmacy.

OpenEMR interface showing the Medical Record Dashboard for CHERYL ADAMS (322). An alert is displayed: "hank.test says New Due Clinical Reminders Response Needed: Possible STEMI Patient (See the Clinical Reminders widget for more details)". The dashboard includes sections for Allergies, Medical Problems, Medications, Prescriptions, Demographics, and Patient Portal / API Access. The Medical Problems section lists conditions such as Tension headache, Chronic obstructive pulmonary disease, unspecified, Elevated blood pressure reading without diagnosis of hypertension, Unspecified essential hypertension, Acute myocardial infarction, unspecified, Conduct disorder, childhood-onset type, Reiter's disease, multiple sites, Pulmonary alveolar microlithiasis, and Drug or chemical induced diabetes mellitus with kidney complications.

Medical Record Dashboard - CHERYL M. ADAMS

Dashboard History Report Documents Transactions Issues Ledger External Data

Allergies Medical Problems Medications Prescriptions Demographics Patient Portal / API Access

Medical Problems:

- Tension headache
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- Pulmonary alveolar microlithiasis
- Drug or chemical induced diabetes mellitus with kidney complications

Alert: hank.test says New Due Clinical Reminders Response Needed: Possible STEMI Patient (See the Clinical Reminders widget for more details)

Cardiology, the cath lab, and pharmacy acknowledge that there is a possible STEMI patient in the ED.

The screenshot shows the OpenEMR interface for a patient named Cheryl Adams. A modal form titled "Response Needed - Possible STEMI Patient" is open, allowing for a response to a clinical reminder. The form contains the following fields:

- Date/Time:** 2024-12-16 20:51:29
- Completed:** YES
- Results/Details:** possible STEMI patient

The modal has "Save" and "Cancel" buttons at the top right and a "Close" button at the bottom right. In the background, the patient's profile is visible, including her name, DOB (1972-02-11), and various clinical reminders. The "Clinical Reminders" section on the right lists several items, all marked as "Past Due":

- Assessment: Exercise
- Response Needed: Possible STEMI Patient
- Measurement: Weight
- Measurement: Mammogram
- Examination: Pap Smear
- Measurement: Blood Pressure
- Treatment: Influenza Vaccine

The Windows taskbar at the bottom shows the system time as 8:51 PM on 12/16/2024.

The clinical reminder indicates "**Not Due**" because the rule was acknowledged within the appropriate timeframe.

OpenEMR

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Calendar Finder Flow Recalls Messages Patient Fees Modules Procedures Admin Reports Miscellaneous Popups

Search by any demographic

CHERYL ADAMS (322) ×

DOB: 1972-02-11 Age: 52

Select Encounter (1) +

Open Encounter: 2024-12-16 (202)

Portal

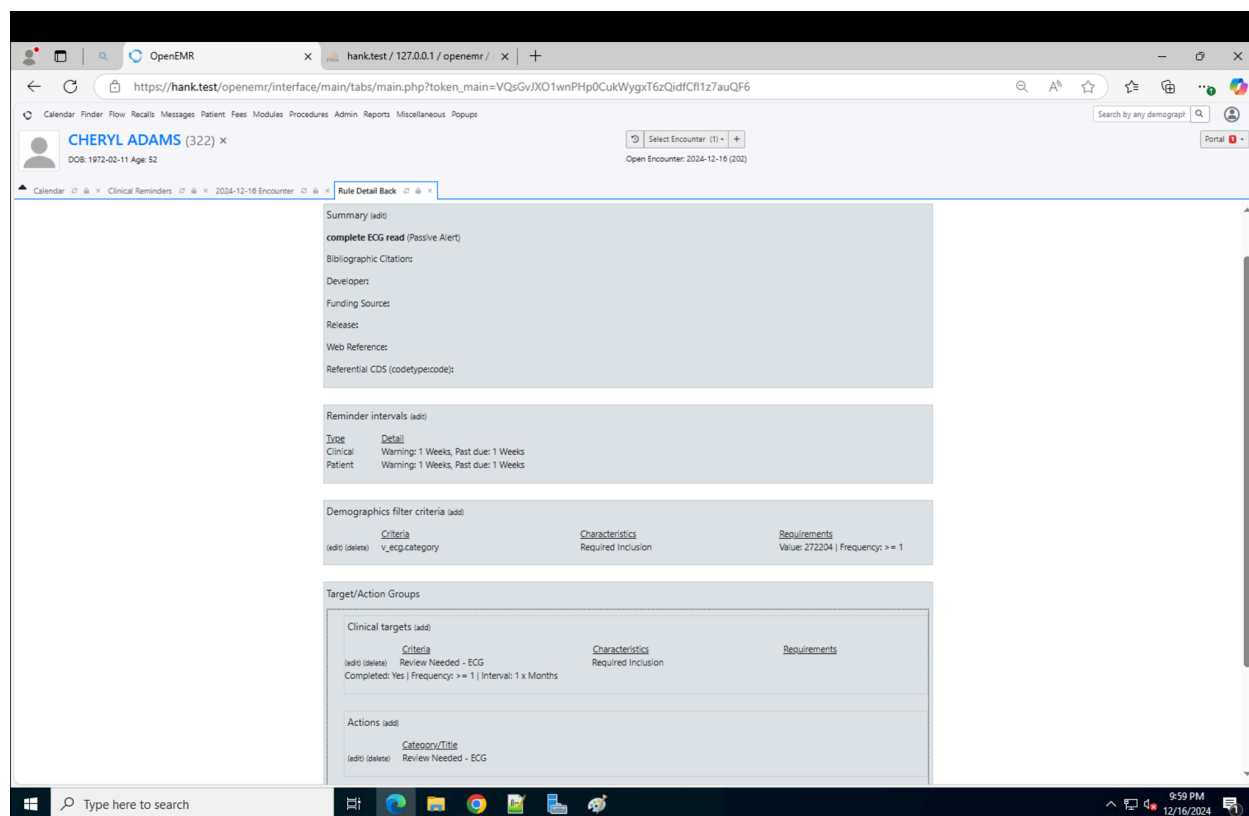
Clinical Reminders

for ADAMS, CHERYL M. Back To Patient

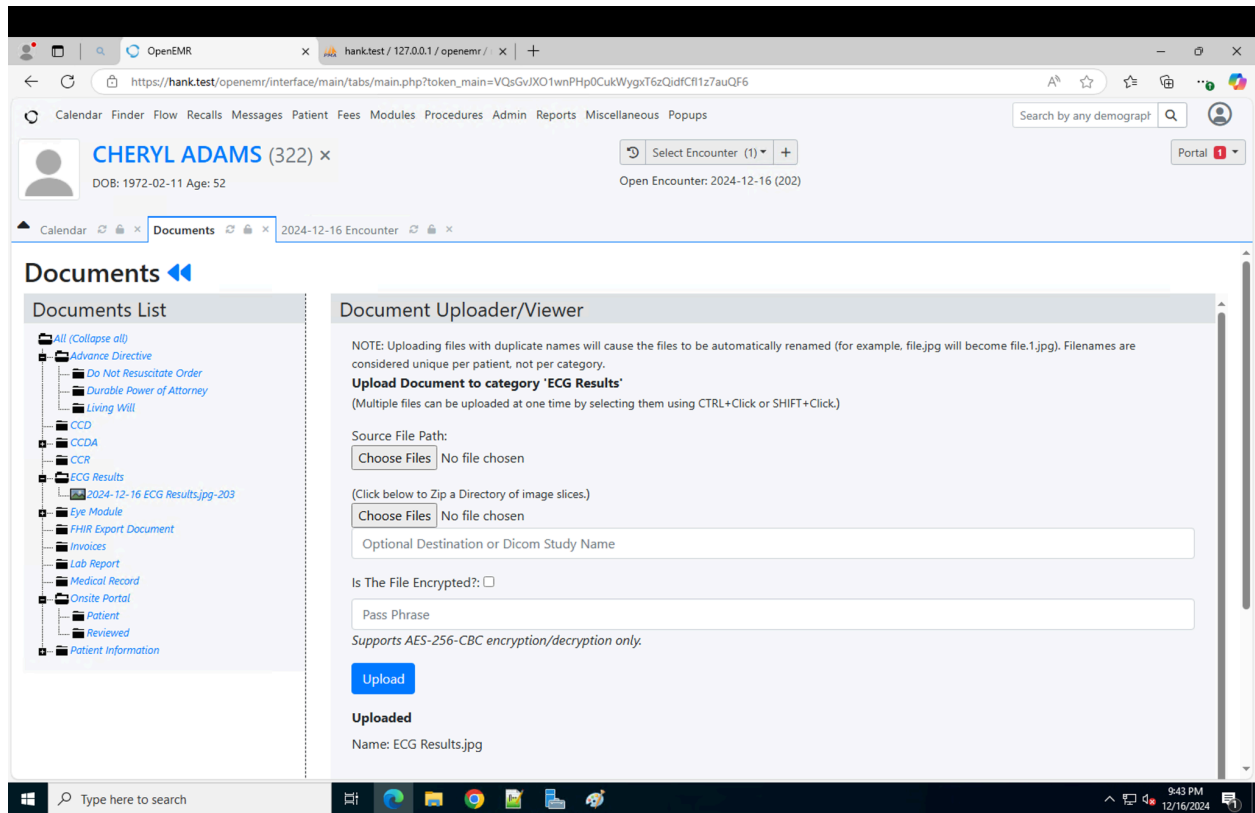
Main Plans Admin

Assessment: Exercise	Past Due
Response Needed: Possible STEMI Patient	Not Due
Measurement: Weight	Past Due
Assessment: Colon Cancer Screening	Not Due
Measurement: Mammogram	Past Due
Examination: Pap Smear	Past Due
Measurement: Blood Pressure	Past Due
Treatment: Influenza Vaccine	Past Due
Assessment: Tobacco	Not Due

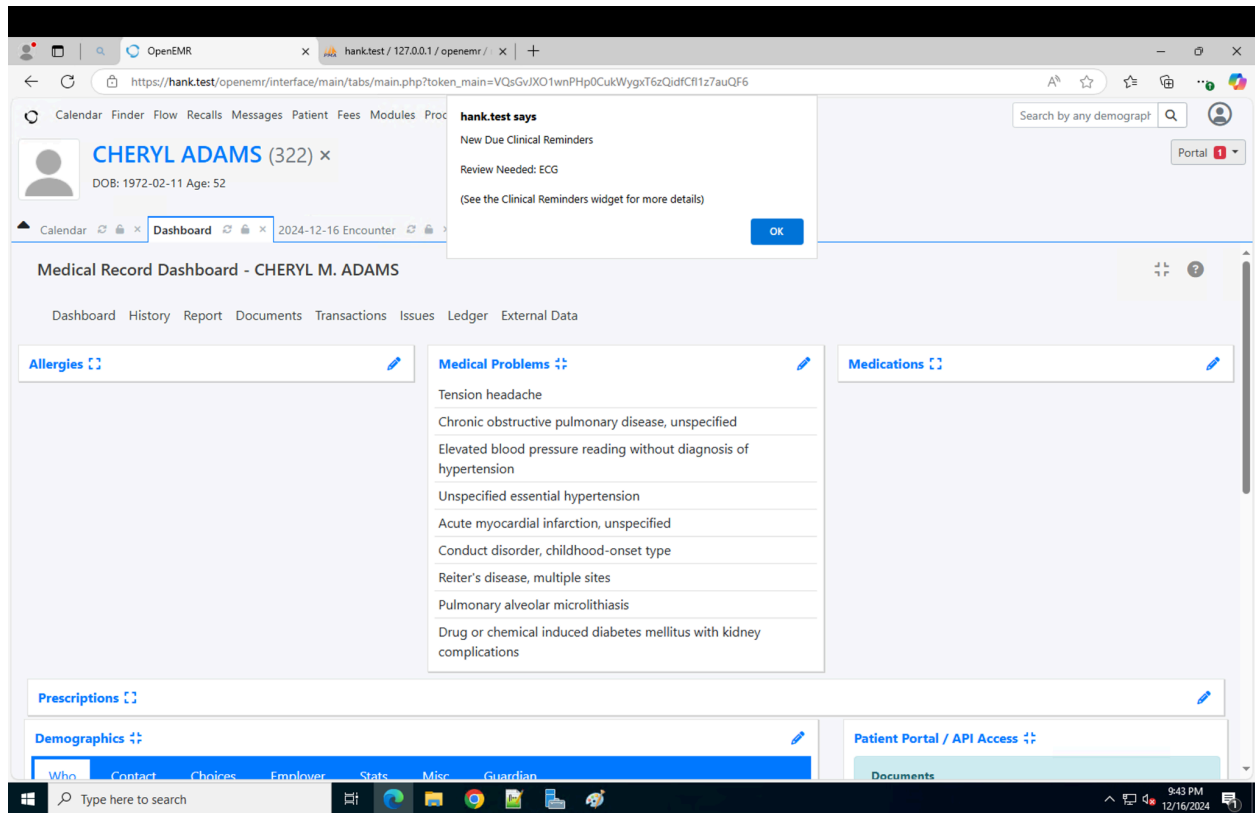
Rule 2: Upon ECG results being uploaded to EMR, send a modal alert to the cardiologist to do the ECG read and confirm STEMI diagnosis.



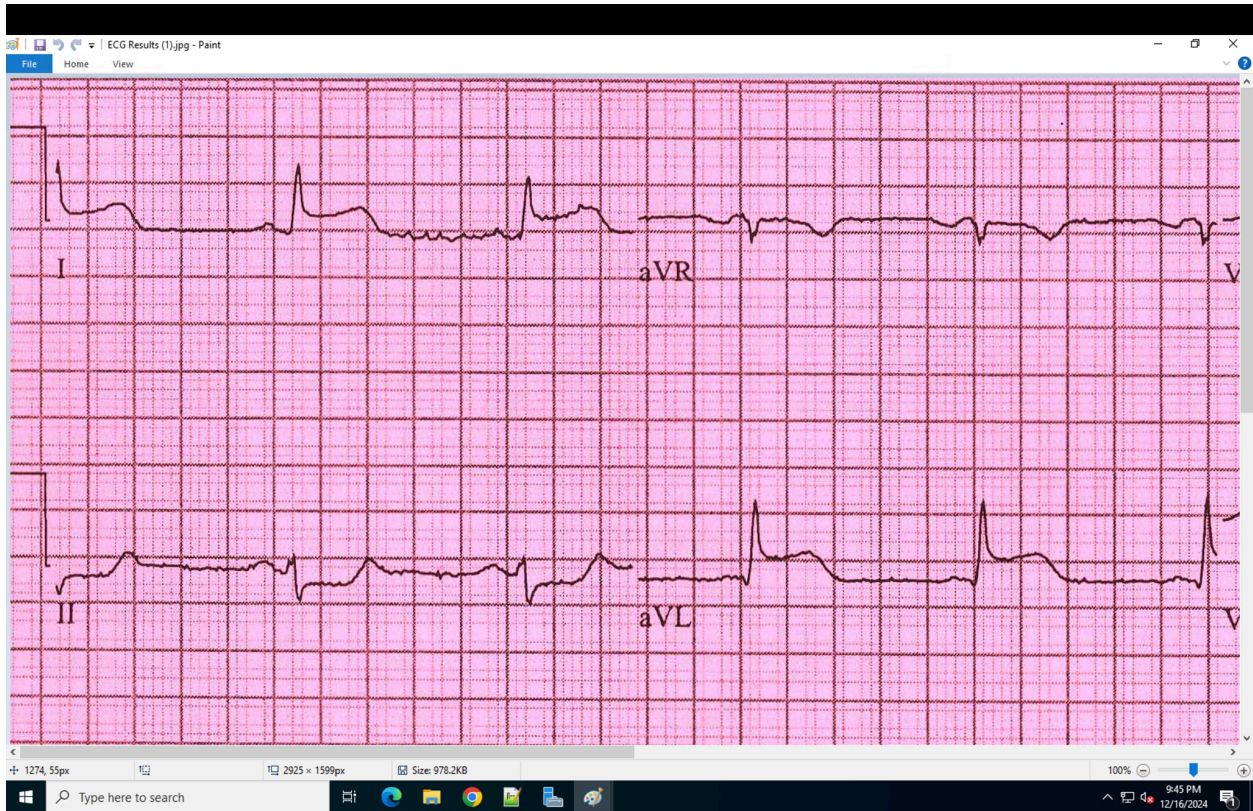
The technician takes the ECG results and uploads them into a newly created document category called **“ECG Results.”**



This triggers an alert for the cardiologist to review the ECG.



The cardiologist navigates to the documents, reviews the results, and confirms that the patient has STEMI.



OpenEMR

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Calendar Finder Flow Recalls Messages Pa

CHERYL ADAMS (322)

DOB: 1972-02-11 Age: 52

Calendar Dashboard 2024

Preferred Name:
Birth Name:
DOB: 1972-02-11
Gender Identity:
External ID: 322
License ID: 607116961
User Defined:
Billing Note:
Previous Names:

Billing Insurance Messages Patient Reminders Disclosures Amendments Labs Vitals

Review Needed - ECG

Date/Time: 2024-12-16 21:48:00

Completed: YES

Results/Details: reviewed. STEMI found

Close

Assessment: Exercise Past Due ?

Review Needed: ECG Past Due ?

Measurement: Weight Past Due ?

Measurement: Mammogram Past Due ?

Examination: Pap Smear Past Due ?

Measurement: Blood Pressure Past Due ?

Treatment: Influenza Vaccine Past Due ?

Recall

Appointments +

Immunizations

Tracks

The clinical reminder indicates **"Not Due"** because the rule was acknowledged within the appropriate timeframe.

The screenshot shows the OpenEMR interface for patient Cheryl Adams (DOB: 1972-02-11, Age: 52). The 'Clinical Reminders' tab is active, showing a list of reminders for the 2024-12-16 encounter. The reminders are categorized by status: Past Due (red) and Not Due (green). The 'Main' tab is selected in the navigation bar.

Reminder	Status
Assessment: Exercise	Past Due
Response Needed: Possible STEMI Patient	Not Due
Review Needed: ECG	Not Due
Activate: Cath Lab	Past Due
Measurement: Weight	Past Due
Assessment: Colon Cancer Screening	Not Due
Measurement: Mammogram	Past Due
Examination: Pap Smear	Past Due
Measurement: Blood Pressure	Past Due
Treatment: Influenza Vaccine	Past Due
Assessment: Tobacco	Not Due

The cardiologist then updates the diagnosis code to STEMI.

The screenshot shows the OpenEMR interface with the 'Add/Edit Issue' window open. The 'Select Codes' tab is active, displaying a search for 'STEMI'. The results list various medical codes, with I21.09 being the most relevant for a STEMI diagnosis.

Code	Description
A44.0	Systemic bartonellosis
C96.0	Multifocal and multisystemic (disseminated) Langerhans-cell histiocytosis
C96.21	Aggressive systemic mastocytosis
C96.5	Multifocal and unisystemic Langerhans-cell histiocytosis
D47.02	Systemic mastocytosis
D72.12	Drug rash with eosinophilia and systemic symptoms syndrome
E85.3	Secondary systemic amyloidosis
G13.1	Other systemic atrophy primarily affecting central nervous system in neoplastic disease
G13.2	Systemic atrophy primarily affecting the central nervous system in myxedema
G13.8	Systemic atrophy primarily affecting central nervous system in other diseases classified elsewhere
I21.01	ST elevation (STEMI) myocardial infarction involving left main coronary artery
I21.02	ST elevation (STEMI) myocardial infarction involving left anterior descending coronary artery
I21.09	ST elevation (STEMI) myocardial infarction involving other coronary artery of anterior wall
I21.11	ST elevation (STEMI) myocardial infarction involving right coronary artery
I21.19	ST elevation (STEMI) myocardial infarction involving other coronary artery of inferior wall

Showing 1 to 15 of 205 entries (filtered from 72,748 total entries)

Rule 3: Upon cardiologist confirming STEMI diagnosis, activate the cath lab and send an alert to the cath lab to start preparing for an incoming STEMI patient.

OpenEMR interface showing the Rule Detail for a STEMI Cath Lab Alert. The patient is CHERYL ADAMS (322), DOB: 1972-02-11, Age: 52. The alert is titled "STEMI Cath Lab Alert (Passive Alert)".

Summary (add)
STEMI Cath Lab Alert (Passive Alert)
Bibliographic Citations
Developer:
Funding Source:
Releaser:
Web References:
Referential CDS (codetypecode):

Reminder intervals (add)
Type: Clinical, Patient
Detail: Warning: 1 Weeks, Past due: 1 Weeks
Warning: 1 Weeks, Past due: 1 Weeks

Demographics filter criteria (add)
Criteria: Diagnosis
Characteristics: Required Inclusion
Requirements: ICD10I21.09

Target/Action Groups
Clinical targets (add)
Criteria: Activate - Cath Lab
Completed: Yes | Frequency: >= 1 | Interval: 1 x Months
Characteristics: Required Inclusion
Requirements:
Actions (add)
Category/Title: Activate - Cath Lab

The cath lab receives an alert to start preparing for an incoming STEMI patient.

OpenEMR interface showing the Medical Record Dashboard for CHERYL M. ADAMS. A notification from hank.test says: "New Due Clinical Reminders, Activate: Cath Lab (See the Clinical Reminders widget for more details)".

Medical Record Dashboard - CHERYL M. ADAMS
Dashboard History Report Documents Transactions Issues Ledger External Data

Allergies [edit]
Medical Problems [edit]
Tension headache
Chronic obstructive pulmonary disease, unspecified
Elevated blood pressure reading without diagnosis of hypertension
Unspecified essential hypertension
Acute myocardial infarction, unspecified
ST elevation (STEMI) myocardial infarction involving other coronary artery of anterior wall
Conduct disorder, childhood-onset type
Reiter's disease, multiple sites
Pulmonary alveolar microlithiasis
Drug or chemical induced diabetes mellitus with kidney complications
Prescriptions [edit]
Medications [edit]

The cath lab begins prepping for the STEMI patient, and the patient is transported for surgery.

The screenshot displays the OpenEMR web interface. A patient record for **CHERYL ADAMS** (DOB: 1972-02-11, Age: 52) is visible in the background. Overlaid on this is a modal window titled "Activate - Cath Lab". The modal contains the following fields:

- Date/Time:** 2024-12-16 21:53:43
- Completed:** YES (selected from a dropdown menu)
- Results/Details:** prepping cath lab

Buttons for "Save", "Cancel", and "Close" are present. The background interface shows various tabs like Calendar, Finder, Flow, Recalls, Messages, and a list of clinical reminders on the right, including "Measurement: Weight", "Measurement: Mammogram", "Examination: Pap Smear", "Measurement: Blood Pressure", and "Treatment: Influenza Vaccine", all marked as "Past Due".

The clinical reminder indicates **"Not Due"** because the rule was acknowledged within the appropriate timeframe.

OpenEMR

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https://hanktest/openemr/interface/main/tabs/main.php?token_main=VQsGvJXO1wnPHpOCukWygxT6zQidfCfl1z7auQF6

Calendar Finder Flow Recalls Messages Patient Fees Modules Procedures Admin Reports Miscellaneous Popups

 **CHERYL ADAMS** (322) ×
DOB: 1972-02-11 Age: 52

Select Encounter (1) +

Open Encounter: 2024-12-16 (202)

Portal 1

Calendar Clinical Reminders 2024-12-16 Encounter

Clinical Reminders

for **ADAMS, CHERYL M.** [Back To Patient](#)

Main Plans Admin

Assessment: Exercise	Past Due
Response Needed: Possible STEMI Patient	Not Due
Review Needed: ECG	Not Due
Activate: Cath Lab	Not Due
Measurement: Weight	Past Due
Assessment: Colon Cancer Screening	Not Due
Measurement: Mammogram	Past Due
Examination: Pap Smear	Past Due
Measurement: Blood Pressure	Past Due
Treatment: Influenza Vaccine	Past Due
Assessment: Tobacco	Not Due

Type here to search

9:57 PM
12/16/2024